Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

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Instructor:	Dr. Muhammad Jaseemuddin

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Assignment/Lab Title:	FastAPI and Containers

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Introduction

This lab report covers the design and implementation of a RESTful HTTP server with the use of FastAPI and a HTTP interface to simulate a rover and mine management system. Communication through the server and operators is done through JSON. The system was containerized through Docker and deployed through Microsoft Azure applications.

Part 1

The server was created through the FastAPI in Python and implements all required endpoints. A route called "/docs" can be accessed at https://coe892lab42022g501177505-f8ftb9ccf5hpgdax.canadacentral-01.azurewebsites.net/docs or locally for testing purposes.

- GET /map, PUT /map: retrieve and resize the mine field grid
- GET /mines, POST /mines, GET/PUT/DELETE /mines/{id}: full CRUD for mines
- GET /rovers, POST /rovers, GET/PUT/DELETE /rovers/{id}: full CRUD for rovers
- POST /rovers/{id}/dispatch: simulates rover movement, mine detection, PIN cracking, and returns the traversal path

The mines and map files are loaded upon startup. Dispatch logic allows for rovers to execute left, right, move, and dig commands, and performs SHA-256 brute force operations. Rovers can be in four different states: Not started, moving, finished, or eliminated.

Part 2

The operator was created as a single HTML file that connects to the server via Fetch API. The interface supports the following functions:

- Viewing the 10x10 map file
- Adding and deleting mines
- Creating rovers with imputed commands
- Viewing dispatch results
- Timestamped log of server responses
- Configurable server URL (supporting both local and azure deployments)

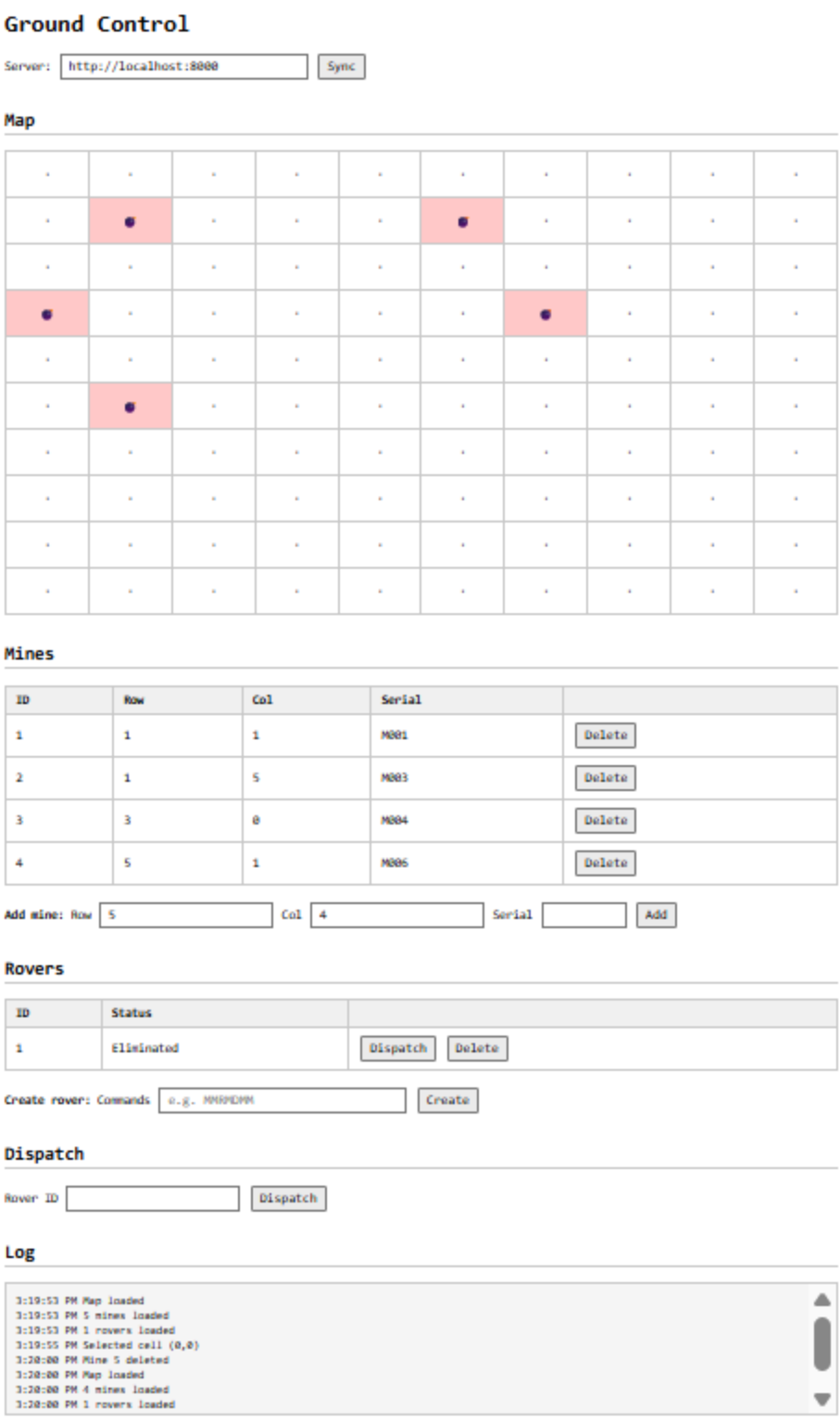

Figure 1. Operator Interface

Part 3

Part 3 of the lab was skipped as it was optional.

Part 4

The server was containerized using Docker and a Dockerfile. The initial container was set up, deployed and pushed to the remote container registry.

```
PS C:\Users\darkt> docker tag lab4-5 COE892Lab42022G501177505.azurecr.io/lab4-5:latest
PS C:\Users\darkt> docker push COE892Lab42022G501177505.azurecr.io/lab4-5:latest
The push refers to repository [COE892Lab42022G501177505.azurecr.io/lab4-5]
e38be7ef3dd0: Pushed
e35b92e2619d: Layer already exists
6944007a5b45: Layer already exists
ad663a8a49bd: Pushed
a4e8b2ba5278: Pushed
7524aba74b8c: Layer already exists
ce370fc2a812: Layer already exists
ec781dee3f47: Layer already exists
91f4ffd50c9c: Pushed
latest: digest: sha256:ab61b501739de9b6b0e83b11ad88455c11874aadc0ec89c93d4ffef566e6e845 size: 856
```

Figure 2. Containerization

Results

Below is a simulation of a rover with commands “MMMM” running and encountering a mine. Upon encountering the mine the rover is eliminated and results are printed in the server log.

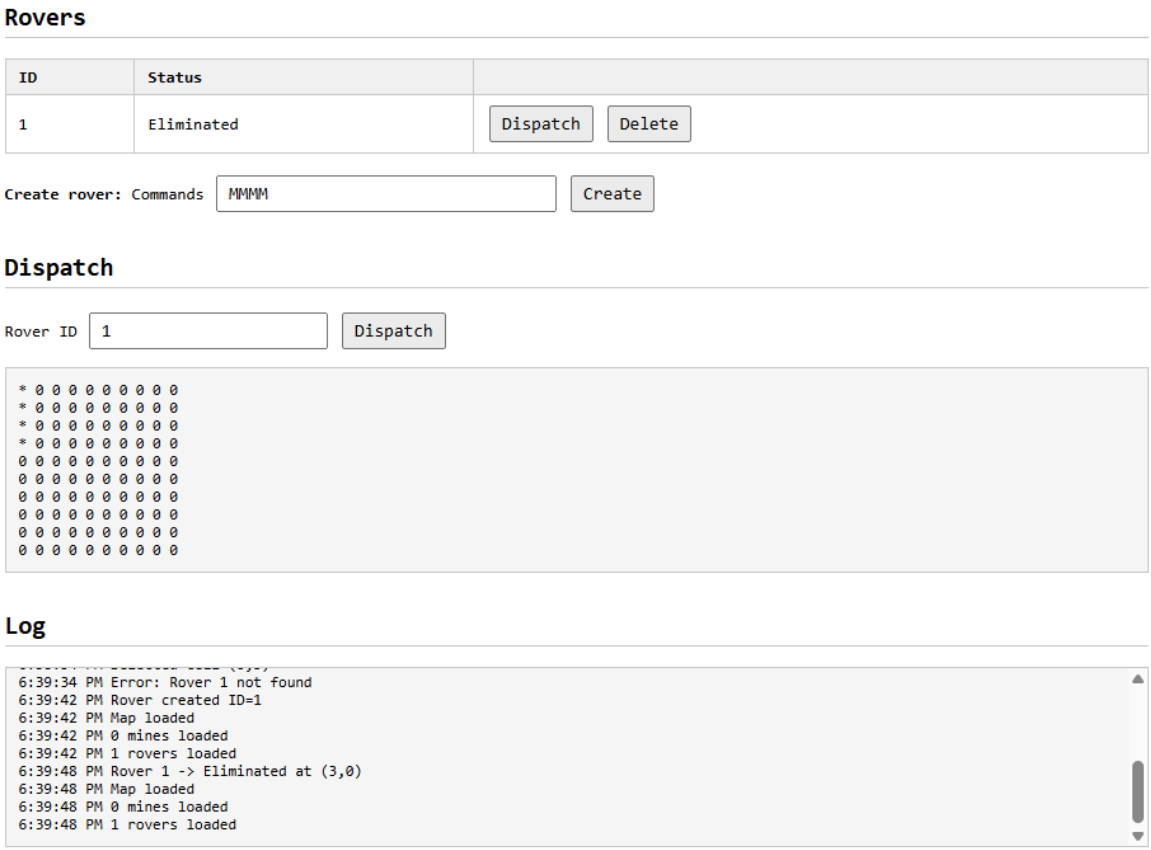

Figure 3. Simulation results

Conclusion

Overall the lab was a success as all three parts of the lab were fully functional. The FastAPI server handles all mapping, mine and rover operations via REST API. The operator was built through HTML and provides a simple user interface connecting to the server. The application was containerized through Docker and deployed to Microsoft Azure making it accessible publicly.